Final Exam Review

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Unit 4

- Compounds/ions that undergo ionization equilibrium.

  Weak Acid:

  Weak Base:

  Cations (that is the conjugate ____ to a weak ______):

  Anions (that is the conjugate ____ to a weak ______):

- % ionization =

Unit 4

- pH of a salt solution
  1. Dissociation: AB (aq) →

  2. Ionization? (Look at their conjugate pairs!)

Cation	**Anion**	**pH?**

Formula	K_a
HIO_3	1.7×10^{-1}
$HC_2H_2O_2Cl$	1.4×10^{-3}
HNO_2	4.6×10^{-4}
HF	3.5×10^{-4}

Formula	K_b
$C_4H_9NH_2$	5.9×10^{-4}
CH_3NH_2	4.5×10^{-4}
NH_3	1.8×10^{-5}

Which one of the following salts, when dissolved in water, produces the solution with the **highest** pH?

A) $NaNO_2$
B) $KClO_4$
C) NH_4Cl
D) LiF

Summary on Le Châtelier's Principle

$$aA + bB \leftrightharpoons cC + dD \qquad K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

New equilibrium shift to the direction to reduce the introduced changes…

Change	Shift Equilibrium?	If So, How?	Change K?
Concentration	_____	increase [reactant], shift to _______ decrease [product], shift to _______ increase [product], shift to _______ decrease [reactant], shift to _______	_____
Volume / Pressure	yes, but only when Δn_____	P↑ or V↓, shift to _______ moles of gases P↑ at fixed V, no effect P↓ or V↑, shift to ______ moles of gases	___
Temperature	____	T↑, shift to the direction ________ heat T↓, shift to the direction ________ heat	_____
Catalysts	____		_____

Your turn!

Consider the following reaction:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) \qquad \Delta H = +\ 235\ \text{kJ}$$

Which disturbance will lead the equilibrium shift to the right?

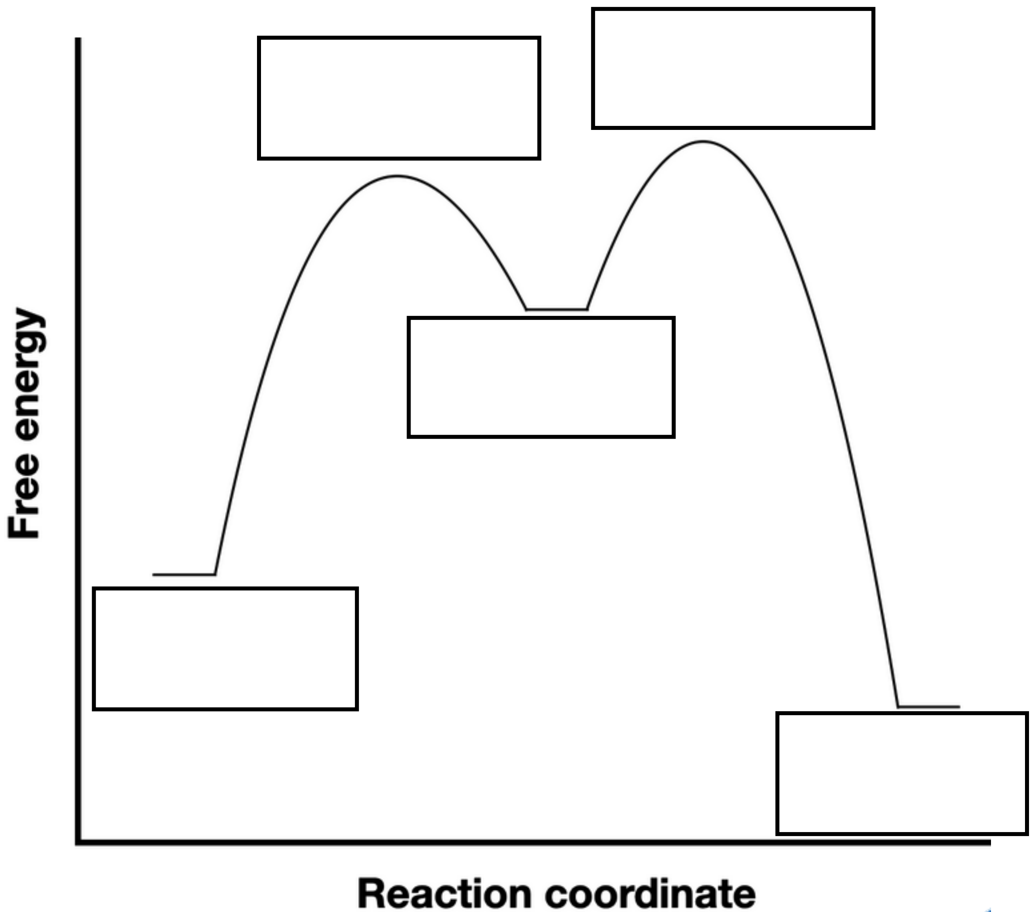
A. Remove Cl_2

B. Decrease the total pressure of the reaction

C. Decrease the temperature of the reaction

Unit 3

1. Interpret a Potential Energy (Reaction Coordinate Diagram) to determine the energy of activation for a reaction and whether the reaction is Exothermic or Endothermic.



Step 1 :

A → B E_{a1} =

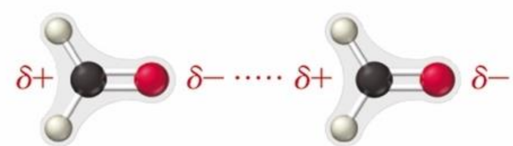
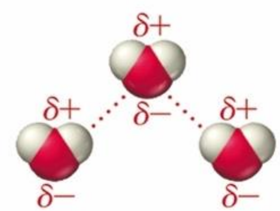
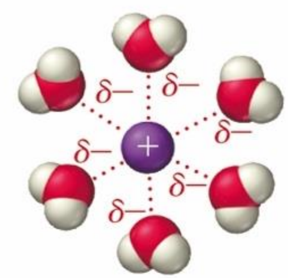
Step 2:

B → C E_{a2} =

ΔH =

Unit 2

1. Determine types and relative strengths of intermolecular forces present in a sample based on the compound's structural formula

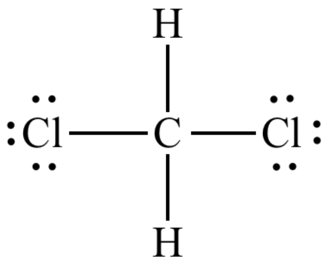
Type	Present In	Molecular Perspective	Strength
			0.05–20+ kJ/mol
			3–20+ kJ/mol
			10–40 kJ/mol
			30–100+ kJ/mol

*The dispersion force can become very strong (as strong and even stronger than the others) for molecules of high molar mass.

Unit 2

1. Determine types and relative strengths of intermolecular forces present in a sample based on the compound's structural formula
2. Relate miscibility of liquids to structure of liquids

What types of intermolecular forces do these compounds have?



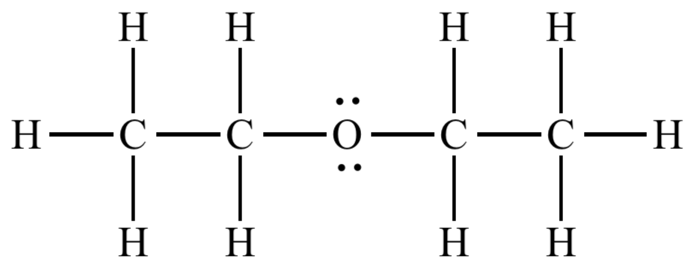
1

Dispersion?

Dipole-dipole?

H bonding?

Ion-dipole?



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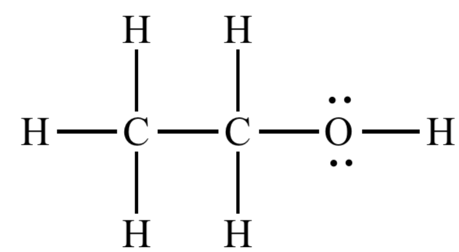
Dispersion?

Dipole-dipole?

H bonding?

Ion-dipole?



3

Dispersion?

Dipole-dipole?

H bonding?

Ion-dipole?

Are they miscible?

Unit 1

1. Understand the concept of spontaneous change and second law of thermodynamics (Entropy of the Universe)
2. Understand the concept of entropy and predict the sign of entropy.
3. Predict the sign of entropy of the system/entropy of the surroundings for a process or chemical reaction.

Entropy (S) increases when …

- The volume ____
- The temperature (average kinetic energy of particles) ____
- the number of molecules ____
- Physical state changes: S(gas)____ S(liquid) ____ S(solids)
- S (pure substances) _____ S(mixtures/solutions)
- the complexity of molecule _______

A. $Br_2 (g) + 3 Cl_2 (g) \rightarrow 2BrCl_3 (g)$ ΔS

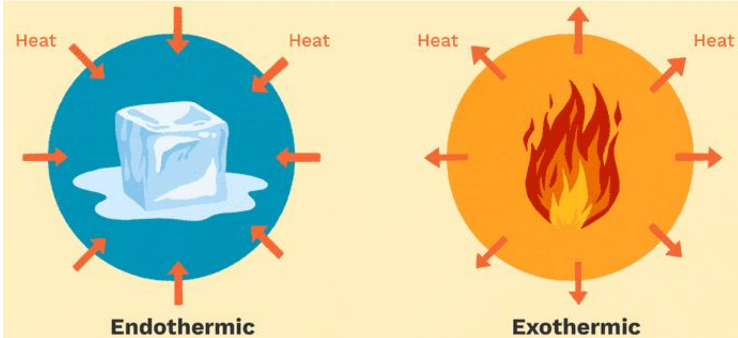
B. $Ni(s) + 2 HCl(aq) \rightarrow H_2(g) + NiCl_2 (aq)$ ΔS

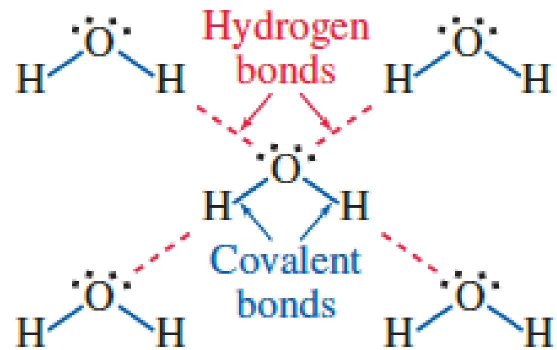
Unit 1

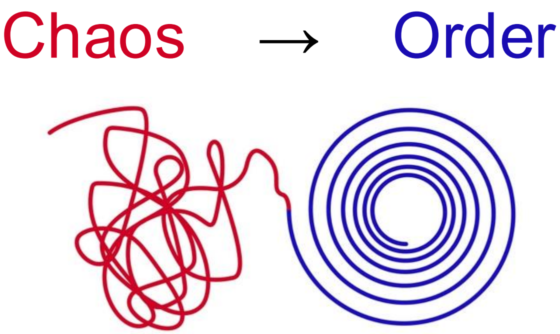
1. Predict spontaneity from ΔH and ΔS.

$$\Delta \boldsymbol{G} =$$

ΔH	ΔS	ΔG	Spontaneous?







We make it!!
Thanks for the fantastic year!